

Documentation for Subject Authority Scripts

Project Scope & Purpose

This suite of Python scripts is developed as an analytical, verification, and audit pipeline to assist the project team in reconciling, replacing, and enriching authority linkages across the dataset.

Initial & Verification Scripts (600_651_version1.ipynb, organization.ipynb):

Serve as specialized testing, verification, and module-level processing scripts. They directly scan existing subject fields (subject_person_600, subject_geo_651, subject_corporate_610), match plain-text entries against individual authority lookup files (LCNAF, GeoNames, VIAF, Wikidata), and route unmatched or ambiguous terms to uncontrolled subject columns for cataloger audit and review.

Updated Main Script (600_610_651_from_keyword_updated.ipynb):

Serves as the integrated pipeline for keyword-level authority resolution. It parses unstructured terms from the keywords column, reconciles them against a multi-authority index (LCNAF, VIAF, GeoNames, Wikidata, and corporate sources) and row-level fallback links, automatically assigns verified entries into subject_person_600, subject_geo_651, or subject_corporate_610, and retains only unmatched keywords in keywords while generating a row-by-row audit log (keyword_authority_log).

Version 1

1. Script: Subject Person and Subject Place Authority Update

The version 1 script (600_651_version1.ipynb) updates subject_person_600 and subject_geo_651. It replaces plain subject names or place names with authority links when a safe match is found. If a name or place cannot be safely matched, the value is moved to an uncontrolled subject column.

```
Controlled output columns:  
subject_person_600 -> LCNAF links only  
subject_geo_651 -> GeoNames links only
```

```
Uncontrolled output columns:  
sub_uncontrolled_person  
sub_uncontrolled_place
```

Input and Output Files

```
MAIN_FILE = ssa_subject_authorities_updated_organization_replaced.xlsx  
CORRECTED_LCNAF_FILE = LCNAF_updated.xlsx  
CORRECTED_GEO_FILE = GeoNames_updated.xlsx  
OUTPUT_FILE = ssa_subject_authorities_updated.xlsx
```

If the local file still uses the older filename containing “corporate,” keep that filename in the code path. The documentation uses “organization” for readability, while the spreadsheet field name remains subject_corporate_610.

Required Main File Columns

```
subject_person_600  
subject_geo_651  
creator_linked_new  
receiver_linked_new  
sender_place_linked_new  
receiver_place_linked_new
```

Processing Logic for subject_person_600

The script loads corrected LCNAF records and builds lookup keys from personStdName and LODLOC_label. It then scans each value in subject_person_600. If a unique LCNAF match is found, the plain name is replaced with an LCNAF anchor. If not found in corrected_lcnaf, the script checks the same row's creator_linked_new and receiver_linked_new columns as a fallback.

```
<p><a href="http://id.loc.gov/authorities/names/[LCNAF_ID]">[LCNAF label]</a> [LC]</p>
```

Person Name Matching and Life-Date Handling

Life dates are removed only during matching so that small date-format differences do not prevent a valid match. The script supports both Last, First and First Last forms. For this project, first-name and last-name matching is acceptable only when it returns one unique authority record.

If the same first and last name could refer to multiple people, such as father/son records with different years, the script does not choose automatically. It treats the case as ambiguous, does not insert a link, and writes the value to the ambiguity log for manual review. For example,

Input subject name: Harrison, James, 1628-1687

Possible normalized match: James Harrison

Safe only when one unique LCNAF record is found.

Example - Update

Old:

subject_person_600

Naylor, James; Benson, Gervase; Nayler, James; Fox, George; Lawson, John

New:

Subject - Person (Linked Data)

```
<p><a href="http://id.loc.gov/authorities/names/n82064043">Naylor, James, 1617?-1660</a>
[LC]</p><p><a href="http://id.loc.gov/authorities/names/n85053694">Benson, Gervase, -
1679</a> [LC]</p><p><a href="http://id.loc.gov/authorities/names/n79065610">Fox, George,
1624-1691</a> [LC]</p>
```

Display:

[Naylor, James, 1617?-1660](#) [LC]

[Benson, Gervase, -1679](#) [LC]

[Fox, George, 1624-1691](#) [LC]

subject_uncontrolled_653_person

Nayler, James; Lawson, John

These uncontrolled terms can be moved to the keywords field.

Processing Logic for subject_geo_651

The script loads corrected GeoNames records and checks each value in subject_geo_651. If a safe match is found, the plain place name is replaced with a GeoNames anchor using the canonical sws URI. If no match is found in corrected_geo, the script checks sender_place_linked_new and receiver_place_linked_new as row-level fallback sources.

```
<p><a href="https://sws.geonames.org/[GeoNames_ID]">[Corrected GeoNames label]</a>
[GeoNames]</p>
```

GeoNames Example

Subject place:

Lancaster Castle, Lancaster, Lancashire, England

Updated GeoNames label:
Lancaster Castle (Lancaster, Lancashire, England)

Added URI format:
[https://sws.geonames.org/\[GeoNames_ID\]](https://sws.geonames.org/[GeoNames_ID])
<https://sws.geonames.org/6287284>

The script replaces the subject-place wording with the corrected GeoNames label and uses the canonical sws.geonames.org URI.

Log Sheets and Review Instructions

Log sheet	What it means
sub_person_log	Shows how each subject_person_600 value was handled.
sub_place_log	Shows how each subject_geo_651 value was handled, including match score.
sub_person_ambig	Multiple person matches were possible.
sub_place_ambig	Multiple place matches were possible.

2. Script: Organization Subject Authority Replacement

This script updates the organization subject field. The actual spreadsheet column is still subject_corporate_610, but the documentation refers to the values as organization subjects because the entries may include churches, societies, institutions, and other bodies.

Controlled field: subject_corporate_610
Uncontrolled field: uncontrolled_subject_corporate_610

Input and Output Files

```
main_file = ssa_viaf_updated.xlsx
corrected_file = organization_updated.xlsx
output_file = ssa_subject_authorities_updated_organization_replaced.xlsx
```

The specific uploaded correction file may have a date in its filename. The filename can be changed in the code path, but the documentation keeps the name generic.

Required Columns

Main file:
subject_corporate_610

Organization correction file:
organizationDesc_original
Wikidata URL
Wikidata Label
LCNAF URL
LCNAF Label
VIAF URL
VIAF Label

Processing Logic

The script builds a replacement dictionary from organizationDesc_original. For each organization, it creates available Wikidata, LCNAF, and VIAF links. If subject_corporate_610 contains a matching organization label, the value is replaced with the authority link or links. If no match is found, the cleaned label is moved to uncontrolled_subject_corporate_610.

URI Normalization

Authority	Script output URI
Wikidata	http://www.wikidata.org/entity/QID
LCNAF	http://id.loc.gov/authorities/names/ID
VIAF	http://viaf.org/viaf/ID

Examples

Wikidata old/correction source:

<https://www.wikidata.org/wiki/Q310439>

Wikidata output:

```
<p><a href="http://www.wikidata.org/entity/Q310439">George Fox (Q310439)</a>
[Wikidata]</p>
```

LCNAF output:

```
<p><a href="http://id.loc.gov/authorities/names/n79065610">[LCNAF Label]</a> [LC]</p>
```

VIAF output:

```
<p><a href="http://viaf.org/viaf/106965214">[VIAF Label]</a> [VIAF]</p>
```

Replacement Behavior

The script handles multiple organization subjects in one cell by splitting on semicolons or newlines. The controlled subject_corporate_610 column keeps only successfully replaced authority links. Unmatched organization names are preserved in uncontrolled_subject_corporate_610 for review.

3. Final Output Rule

After these scripts run, the controlled subject fields should contain only authority links. Values that could not be safely matched are not discarded; they are moved to uncontrolled columns for manual review.

```
subject_person_600 -> LCNAF links only
```

```
subject_geo_651 -> GeoNames links only
```

```
subject_corporate_610 -> organization authority links only
```

```
sub_uncontrolled_person -> unmatched person names
```

```
sub_uncontrolled_place -> unmatched place names
```

```
uncontrolled_subject_corporate_610 -> unmatched organization names
```

Updated Script Documentation

Script File Name

600_610_651_from_keyword_updated.ipynb

Overview & Core Functionality

This updated script automates the processing, normalization, and authority reconciliation of the **keywords** column in the primary dataset.

Instead of operating directly on existing subject fields, the script parses individual terms from the keywords column and checks them against targeted entity authority files:

1. **Person Terms:** Matches terms against person authority files and moves successfully identified entities into **subject_person_600** as HTML-formatted authority links.
2. **Place Terms:** Matches terms against geographic authority files and moves identified entities into **subject_geo_651** as HTML-formatted authority links.
3. **Organization Terms:** Matches terms against corporate/organization authority lists and moves identified entities into **subject_corporate_610** as HTML-formatted authority links.
4. **Row-Level Fallback Reconciliation:** If a term is not found in the external authority spreadsheets, the script performs a row-level secondary lookup against existing linked fields in the same row (**creator_linked_new** / **receiver_linked_new** for persons, and **sender_place_linked_new** / **receiver_place_linked_new** for places).
5. **Unmatched Keyword Retention:** Successfully matched terms are **removed** from keywords. Only terms that cannot be safely reconciled against any authority source or fallback column remain in keywords for subsequent manual cataloger review.

Controlled Output Columns & Supported Authorities

- **subject_person_600** (Controlled Persons):
 - **Supported Authorities:** LCNAF, VIAF, Wikidata
 - **Output Format:** <p>[Label] [LC/VIAF/Wikidata]</p>
- **subject_geo_651** (Controlled Places):
 - **Supported Authorities:** GeoNames, Wikidata
 - **Output Format:** <p>[Label] [GeoNames/Wikidata]</p>
- **subject_corporate_610** (Controlled Organizations):
 - **Supported Authorities:** LCNAF, VIAF, Wikidata
 - **Output Format:** <p>[Label] [LC/VIAF/Wikidata]</p>
- **keywords** (Unmatched Terms):
 - Retains only plain-text keywords that could not be matched and migrated.

Input and Output File Configuration

- **Main Dataset (MAIN_FILE):** franckenachlass_7.xls_Wed_Sep_09_07_12_33_2026part_1.xlsx
- **Authority Files:**
 - LCNAF_updated.xlsx (LCNAF Person)
 - VIAF_updated.xlsx (VIAF Person)
 - Wikidata_person_updated.xlsx (Wikidata Person)
 - GeoNames_updated.xlsx (GeoNames Place)
 - Wikidata_place_updated.xlsx (Wikidata Place)
 - organization_updated.xlsx (Organization / Corporate)

- **Output Dataset (OUTPUT_FILE):**
franckenachlass_8.xls_Wed_Sep_09_07_12_33_2026part_1.xlsx

Audit Log Sheet

Upon execution, the script appends an audit sheet to the output workbook:

- **keyword_authority_log:** Contains row-by-row tracking for every extracted keyword term, detailing the assigned entity category, authority source or fallback used, URIs, labels, or its status as an unmatched term.

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